

Neural Network Prediction Using The Stock Market

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Abstract

Neural Networks have seen use in predicting the stock market, owing to their ability to model complex high dimensional non-linear relationships.

This is not that. This is the opposite of that.

In this paper we construct, using only option contracts, a financial exotic whose value is equal to that of a Neural Network's output. Specifically, let there be some number of assets with value $V_i(t)$ each. Then, at a set time T in the future, the exotic is worth $\text{NN}(V_0(T), V_1(T), \dots)$. This method allows for Neural Networks composed of fully connected layers, ReLU activations, and other basic architectures. Finally, we convert a Neural Network trained for solving the XOR problem into an exotic, which we term the "eXORTic".

Project page: <https://javalim.com/eXORTic>

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1 Artificial Neural Networks

In its most basic form, Neural Networks have two components. Neurons, which store and process signals, and directed edges between neurons, which represent weighted connections. For a non-input neuron, its value is given by the weighted sum of all neurons pointing into it, potentially added with a constant known as the bias. This sum is then passed into a non-linear function, known as the activation function. Activation functions introduce nonlinearities into the network, giving it high expressive power.

The simplest case involves a single neuron, y , which has vector of input neurons $\mathbf{x}$. The weighted sum of these inputs, added to a bias, would be written as $\mathbf{w} \cdot \mathbf{x} + b$,

for weight vector $\mathbf{w}$ and bias b . Finally, we apply the activation function, giving $y = \phi(\mathbf{w} \cdot \mathbf{x} + b)$ for activation function ϕ . When grouping into layers of neurons, the matrix form $\mathbf{y} = \phi(\mathbf{W}\mathbf{x} + \mathbf{b})$ is preferred. Extending this structure by adding more layers increases the depth of the network, allowing it to model more complex patterns and relationships in data.

In recent literature, a common choice of activation function is the ReLU function, given by $\text{ReLU}(x) = \max(0, x)$, elementwise.

2 Financial Derivatives

Derivatives are financial instruments whose value depends on (or, *derives* from) the value of an underlying asset. For instance, a “call option” on a gold bar (the “underlying”) is a contract that gives the holder the right, but not the obligation, to buy a gold bar for a predetermined price (the “strike price”), at a predetermined time (the “expiration date”). When the true price of gold is higher than the strike price, there is then value in owning this contract.

2.1 Call Options

In particular, let $S(t)$ be the price of an underlying asset. Let there also be a call option on this asset, with strike price K , and expiration date T . The value of the option at expiry can be split into two cases. Case 1, $S(T) \geq K$. The holder of the option can exercise their right, and buy the asset for just K and immediately sell it off again for $S(T)$, making a profit of $S(T) - K$. Case 2, $S(T) \leq K$. It is not logical to exercise the option here, as we would be purchasing the asset at a higher price. As we are not obligated to do so, we don’t. The option thus has a value of 0, and is said to have expired worthless.

Combining the cases, we may write that the option has value at expiry $V(T) = (S(T) - K)^+ = \max(0, S(T) - K)$.

For the other party, their payoff is exactly the negative of this value, as they would be selling an asset for less than it is worth. This is equivalent to having a negative quantity of the option.

We note in passing that, in the real world, money is paid to initially create this option (the “premium”). Thus we start with a loss, and profit more as the underlying gains value. On the other hand, the seller of the option starts with a profit, and loses more as the underlying gains value. We do not consider the premium in this paper, as fairly pricing options is far out of scope. We will only work with the intrinsic value of the option, not considering how much it cost to create it.

2.2 Basket Options

Now, suppose instead of a single asset, we would like to place a bet on a group of assets (perhaps stocks of companies we find promising). Someone could construct a portfolio of these assets, then sell an option contract on this portfolio. This combined action is also known as a basket option. As they are less common, they may be classified as “exotic derivatives”, “exotic options”, or simply, “exotics”.

The underlying is now a weighted average of assets, $\sum_i w_i S_i(t)$. As such, the option at expiry is now worth $V(T) = \max(0, \sum_i w_i S_i(t) - K)$. Compacting into a vector, $V(T) = \max(0, \mathbf{w} \cdot \mathbf{s}(T) - K)$

3 Hmm...

Hmm...

4 The Punchline

Evidently, there is a natural correspondence between basic fully-connected, ReLU activated neural networks, and hierarchies of basket options. In particular, a single neuron with inputs is analogous to a basket option on some underlying assets. For deeper layers, these underlying assets are themselves basket options.

To be explicit, let us consider a single neuron. If it is an input neuron, we assign an underlying asset to it, such as a stock. The value of the neuron y corresponds to the value of the asset S . Note that stock prices never fall into the negatives. If you require negative numbers to be representable, one approach would be to construct a portfolio consisting of the asset and a negative amount of cash (debt). The total value of this portfolio can then be negative.

A hidden or output layer neuron with ReLU activation corresponds to a basket option on the assets associated with its input neurons. The weights of the basket are equal to the weights between the neurons. Accounting for the bias term, we also add $\$b$ to the basket. Finally, we set the strike price to $\$0$. Our final expression for the value of the basket option at expiry is $(\mathbf{w} \cdot \mathbf{x} + b - 0)^+ = \text{ReLU}(\mathbf{w} \cdot \mathbf{x} + b)$ for corresponding input neuron assets $\mathbf{x}$. (For bias equal to $-b$, one could also just set the strike price to $\$b$). This is exactly the value of the corresponding neuron.

We note that this method is only for inference of the network. Basic convolutional and recurrent networks can also be converted in the same manner. Though they have special considerations during training, if we are able to unroll their loops during inference, we can treat them as feedforward networks.

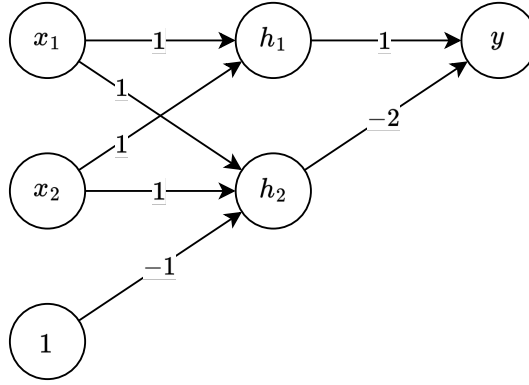
5 The eXORtic

5.1 The XOR problem

The XOR problem is a classic problem in machine learning. The XOR ($\oplus$) operation is defined as

x_1	x_2	$x_1 \oplus x_2$
0	0	0
0	1	1
1	0	1
1	1	0

It serves as a classic example of a problem that is not linearly separable. It is, however, solvable using the following neural network.



h_1 , h_2 , and y all use ReLU activations. The correctness can be verified below.

x_1	x_2	$x_1 + x_2$	h_1	$x_1 + x_2 - 1$	h_2	$h_1 - 2h_2$	y
0	0	0	0	-1	0	0	0
0	1	1	1	0	0	1	1
1	0	1	1	0	0	1	1
1	1	2	2	1	1	0	0

5.2 The eXORtic contracts

Using the method outlined in section 5, we may convert into a set of option contracts.

First, suppose there exist two assets A_1 and A_2 , currently worth \$0, but may be worth \$1 tomorrow.

Now for the first layer, create the following contracts¹

Official contract C_1 !!

This contract entitles the holder the right, but not the obligation, to purchase one quantity of A_1 and one quantity of A_2 for \$0 in total from the seller. This right can only be exercised at exactly XYZ day and time.

Official contract C_2 !!

This contract entitles the holder the right, but not the obligation, to purchase one quantity of A_1 and one quantity of A_2 for \$1 in total from the seller. This right can only be exercised at exactly XYZ day and time.

And for the output layer,

¹I am not a lawyer nor a financial consultant dont do this but if you do it you can't sue me but if you make lots of money can I have some

Official contract “eXORtic” !!

This contract entitles the holder the right, but not the obligation, to purchase one quantity of C_1 and assume responsibility of the seller in two quantities of C_2 for \$0 in total from the seller. This right can only be exercised at exactly XYZ day and time.

Then the eXORtic contract is worth exactly $A_1 \oplus A_2$ at expiry.

To verify this, let us step through the pricing logic when the contracts expire tomorrow.

5.3 Walkthrough: $A_1 = 0, A_2 = 0$

The net value of one A_1 and one A_2 is $\$0 + \$0 = \$0$.

The option contract C_1 allows us to buy one of both for the same price, \$0. As such, it's worth \$0.

The option contract C_2 allows us to buy one of both for an increased price, \$1. Even if we wanted to buy A_1 and A_2 , we would just buy it for the market price, rather than at the strike price. We are not obligated to exercise the option, and hence don't. The option expires worthless, \$0. To the seller, the cost to be responsible for maintaining this option is also \$0, as is certain that no rational holder would exercise the option.

The option contract “eXORtic” allows us to buy one C_1 , and also take responsibility for two C_2 contracts for a price of \$0. C_1 is worth \$0, and C_2 costs \$0 to be responsible for. Hence the overall value is \$0. The option allows us to buy it for the same price, \$0. As such, it's worth \$0 to the holder.

$$A_1 \oplus A_2 = 0 \oplus 0 = 0 = \text{eXORtic}.$$

5.4 Walkthrough: $A_1 = 1, A_2 = 0$ or $A_1 = 0, A_2 = 1$

The net value of one A_1 and one A_2 is $\$1 + \$0 = \$1$.

The option contract C_1 allows us to buy one of both for just \$0. As such, it's worth $\$1 - \$0 = \$1$.

The option contract C_2 allows us to buy one of both for the same price, \$1. There's no difference, as such it's worth \$0 to the holder. Suppose for sake of argument that the holder still wants to exercise it. The seller can spend \$1 to buy one A_1 and one A_2 , then sell it to the holder to recover the \$1. Thus, it also costs \$0 for the seller to be responsible for it.

The option contract “eXORtic” allows us to buy one C_1 , and also take responsibility for two C_2 contracts for a price of \$0. C_1 is worth \$1, and C_2 costs \$0 to be responsible for. Hence the overall value is \$1, which the option allows us to buy for just \$0. Thus, it is worth \$1.

$$A_1 \oplus A_2 = 0 \oplus 1 = 1 = \text{eXORtic}.$$

5.5 Walkthrough: $A_1 = 1, A_2 = 1$

The net value of one A_1 and one A_2 is $\$1 + \$1 = \$2$.

The option contract C_1 allows us to buy one of both for just \$0. As such, it's worth $\$2 - \$0 = \$2$.

The option contract C_2 allows us to buy one of both for just \$1. As such, it's worth $\$2 - \$1 = \$1$. For the seller, they would need to sell $A_1 + A_2$ for \$1 less than it should be worth. As such, it costs \$1 to be responsible for.

The option contract "eXORtic" allows us to buy one C_1 , and also take responsibility for two C_2 contracts for a price of \$0. C_1 is worth \$2, and C_2 costs \$1 to be responsible for. Hence the overall value is $\$2 - 2 \times \$1 = \$0$. The option allows us to buy it for the same price, \$0. As such, it's worth \$0 to the holder.

$$A_1 \oplus A_2 = 1 \oplus 1 = 0 = \text{eXORtic}.$$

6 Practical Notes

6.1 Negative Numbers

Negative numbers are not so easily interpreted in reference to quantities of a physical asset. In general, however, there does exist well defined meaning for them.

In particular, owning -1 stock is known as "shorting" the stock. You have borrowed the stock from someone, and sold it off. The hope is that the future price will go down, allowing you to buy back the stock and return it to the owner, pocketing the difference. Doing the math, you'd see that your profit (or loss) is indeed the negative of the change in the stock's price (ignoring any fees).

As discussed, owning -1 of an option is the same as being on the other side of the trade, the seller.

6.2 Non-integers

Owning a fractional amount of a stock is also challenging. Unfortunately for us, this will be common, as weights are rarely nice integers. One solution is to scale up (to several orders of magnitude) all relevant quantities, then round to the nearest integer. An easier solution would be to cash-settle the options. Instead of the seller actually giving you some number of stock, you agree to receive the cash equivalent. This is actually preferred in this scenario, as we're only concerned in the cash value of the options during the computation of the network's output.

7 Conclusion

In conclusion, we have demonstrated how to represent a standard feedforward ReLU activated neural network using hierarchies of basket options. While the eXORtic is just a toy example, we hope we will one day see trade bootstrapping, wherein trading models run on stock markets run on trading models run on stock markets run on tra